

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MARKING SCHEME

NOVEMBER 2015

CHEMISTRY

9189/5

1 (a) Table 1.1

Give one mark for recording all weighings to at least 2 decimal places. *Sum* *AP* [11]

Give one mark for correct subtraction. *Sub* [11]

(b) Table 1.2

Give one mark for recording all burette readings to 2 d.p, ticked titres and working. *average titre* *Completing Summary correctly* [11]

Give one mark for correct subtraction not more than one zero starts. *1* [11]

Give one mark for sufficient titres *Co* [11]

Accuracy

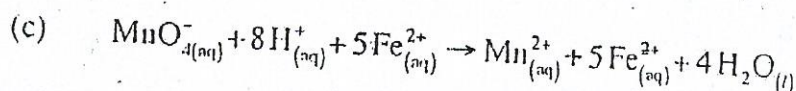
Calculate the ratio $\frac{\text{mass in FA 1}}{\text{volume of FA 2}}$

(Supervisor Titre - 10x Mass FA 1) - (Candidate Titre - 10x Mass FA 1)

For the candidate and assign accuracy marks by comparing this ratio with that obtained for the supervisor. Work to be applied using the table below.

Mark	accuracy marks
≤ 0.05	14
≤ 0.10	13
≤ 0.15	12
≤ 0.20	11
≤ 0.25	10
≤ 0.30	9
≤ 0.50	8
≤ 0.60	7
≤ 0.70	6
≤ 0.85	5
≤ 0.95	4
≤ 1.05	3
≤ 1.40	2
≤ 2.00	1
> 2.00	0

Range used/cm ³	spread penalty
≤ 0.20	0
0.20+ to 0.25	1
0.25+ to 0.30	2
0.30+ to 0.35	3
0.35+ to 0.40	4
0.40+ to 0.50	5
0.50+ to 0.60	6
0.60+ to 0.70	7
0.70+ to 0.80	8
0.80+ to 1.00	9
1.00+ to 1.50	10
1.50+ to 2.00	11
> 2.00	12



(d) Number of moles $\frac{\text{Volume of FA 2} \times 0.001 \text{ moles}}{1000}$ [11]

(e) ans (d) x 5 [11]

$$(f) \text{ Ans (e)} \times \frac{250}{25} \times 55.8 /$$

$$\text{ans (e)} \times 10 \times 55.8 \text{ --- (1)}$$

dividing by 3 for the pill

$$\text{ans (e)} \times 55.8 \times \frac{1}{3} / \text{--- (1)}$$

$$\text{ans (e)} \times 186 \text{ without units (R)}$$

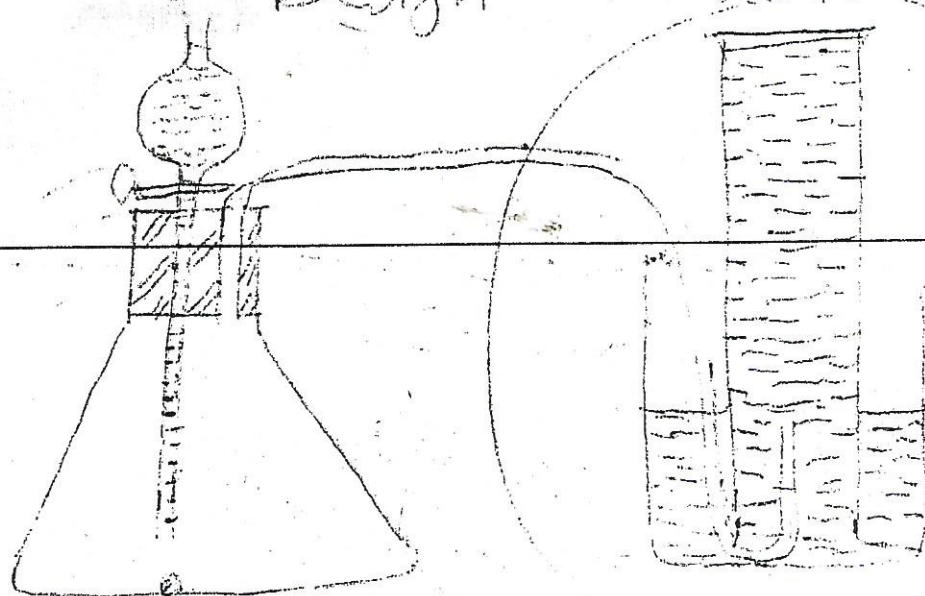
$$(g) \frac{\text{ans (f)} \times 100\%}{m(\text{FAI})} = \frac{3 \times \text{ans (f)} \times 100\%}{m(\text{FAI})}$$

$$\frac{\text{Ans (e)} \times \frac{250}{25} \times 55.8 \times 100\%}{m(\text{FAI})} \text{ (1)}$$

(h). The $\text{Mn}^{2+}/\text{MnO}_4^-$ reacted with the (double bond) of fumarate to form a diol (1)

(i) Colour changes from purple (MnO_4^-) to (pale) pink (Mn^{2+})
 / colourless (Fe^{2+}) to (pale) pink
 / pale green (Fe^{3+}) to (pale) pink (1)

Design No 2015



step 1: Wrap the Na in absorbent paper to 1
remove oil.

step 2: Weigh a small piece of Na ($\leq 0.1g$) 1

step 3: mixing of Na and H_2O / (A) the diagram 1
showing air tight stopper.

step 4: Record the volume of H_2 produced (V) 1
until ALL $m(Na)$ has reacted / No more gas 1
bubbles are seen.

point 5: calculate $n(H_2)$ as $\frac{V}{24000}$ mol. 1

point 6: Quote the rxn ratio Na : H_2 as 2 : 1 1



point 7: Calculate $n(Na)$ as $2n(H_2) = \frac{2V}{24000}$ mol 1

point 8: Calculate $A_r(Na)$ as:

$$\frac{2V}{24000} = \frac{m(Na)}{A_r(Na)} \Rightarrow A_r(Na) = \frac{24000 m(Na)}{V}$$

Σ